In this example, when the user clicks the word Home, the browser would jump to the home.html page.

When pages within the same site are linked, document relative URLs can be used. For example, when the page that is being linked is within the same folder, they can be linked as follows:

```
<a href=”home.html”>Home</a>

<a href=”services.html”>Services</a>

<a href=”products.html”>Products</a>
```

When another site on the web needs to be linked, an absolute URL must be specified. For example,

```
<a href=”http://www.nie.lk/index.htm”>National Institute of Education</a>
```

It will take the user to the home page of the National Institute of Education.

Competency Level 10.5: Uses Style sheet to change the appearance of web pages

Learning Outcomes:

- Briefly explains style sheet and its usage
- Uses the comments and correct syntax in CSS
- Uses appropriate selectors to select elements in CSS
- Inserts CSS in HTML web pages to improve the appearance
- Applies various CSS formatting in HTML web pages to improve the appearance

Style sheets to change the appearance of web pages

Cascading Style Sheets (CSS) provide the look and feel for the HTML content. CSS enables the designer to change how the text, images, and links appearon a web page quickly and easily. Using the CSS, the designer has the freedom of styling the web page independent of the content using a presentation layer where you can make global formatting options. The cascading part of the CSS tells how the elements on a web page are affected by the CSS rules. With CSS, web page styles are made up of one or more style rules.

A CSS rule has three components: the selector, the properties and their values. For example, the following is an HTML rule,

```
h1{color : red;}
```

Where h1 is the selector, color is the property and red is the value assigned for color. Multiple properties and values can be assigned for a single selector, enclosed within curly braces and separated by semicolons. For example, the following CSS rule says that the heading selector h1 has three properties and their corresponding values:

```
h1{
        color : blue;
        margin : 0 ;
        padding : 5px;
}
```

CSS Placement

CSS rules can be integrated into an HTML page in a number of ways: as an external style sheet, embedded within the HTML page itself, and inline as an attribute within the tag.

External style sheets

Theseare used to provide a consistent look-and-feel to any number of related pages, up to and including an entire website. An external style sheet is connected to an HTML page in one of twoways:either with a <link> tag or with a @import directive within a <style> tag. For example, say you wanted to include the CSS rules written in a file called main.css. The <link> syntax would look like this:

```
<link href="styles/main.css" type="text/CSS" rel="stylesheet" />
```

The href attribute provides the path to the external style sheet, and type specifies the kind of document the browser can expect. The relationship of the HTML page to the linked file is defined by the rel attribute; the two possible values are stylesheet and alternate stylesheet. If you wanted to use the @import syntax, you would write code like this:

```
<style>  @import { URL("styles/main.css"); } </style>
```

Notice that @import is actually a CSS rule with the single URL property, written somewhat differently from standard CSS declarations. When used with an HTML page, the @import rule must be within a <style> tag.

External style sheets have the tremendous advantage of being able to affect multiple HTML pages simultaneously. Change any CSS rule, save the external style sheet, publish it, and immediately the modification can be seen by any site visitor to any of the associated pages. You can see why external style sheets are widely used by web designers across the industry.

Internal or embeddedstyles

CSS rules can also be included in the <head> section of an HTML page. CSS rules are embedded through use of the <style> tag, like this:

<style type="text/css"> body {margin: 0; padding: 0; background-color: white ;} h1, h2, h3, h4 {color: red; margin: 0; padding: 5px ;} </style>

Inline

An inline style is applied by using the style attribute within an HTML tag. For example, if you want to colour a<h1> tag red with in an inline style, your code would look this:

<h1 style="color: red;"> Important Message Ahead</h1>

CSS selectors

In CSS there are four basic types of selectors:

- Tags: An HTML tag can serve as a CSS selector. The use of HTML tags as a CSS selector is very straightforward. When an HTML tag, such as <p>, is defined as a selector with CSS, all <p> tags are immediately affected unless another CSS style overrules it.

- IDs: Typically ID selectors are intended to be used once per HTML page. An ID can be defined by using a leading hash (#) symboland can then be applied in an HTML tag using ID attribute. For example, consider the following ID definition and its application.

  #header {width: 960px;}
  ** **

- Classes:A class is another custom selector, which can be used as many times as needed on a web page. To define a class selector a "." symbol is used at the beginning.

  .legalNotice { font-size: small; }

  To apply the class selector to an HTML tag, use the class attribute:

  ** **

  The names of classes and IDs must begin with a letter and not contain any white spaces or other special characters. Similarly, classes and IDs are case-sensitive. In other words, .firstParagraph is not the same as.FirstParagraph.

- Compound:Tags, IDs, and classes can be combined to create a compound selector, which pinpoints a particular section of the page.

Validating CSS Rules

It is essential to ensure that the page that has been designed is error-free and the CSS is syntax error-free. CSS validators can be used for this purpose, which ensures that the CSS is accurate

and does not have any unsupported selectors, properties, or values. The W3C hosts a CSS validation service and it is hosted at **http://jigsaw.w3.org/css-validator**

Styling Text with CSS

The font family property lists a series of fonts that can be used, therefore even if one font is not available on the computer on which the browser is running the other font will be taken automatically by the browser. For example, the declaration,

font-family: Arial, Helvetica, sans-serif;

It makes the browser to first look for the font Arial, if it is not available, look for Helvetica and if that too is not available look for sans-serif.

The size of the font for text on the Web is determined by the font-size property. You can use a named, relative measurement such as large or small like this:

h1 { font-size: x-large; }

Colours for text can be set by the CSS colour property by using

h1, h2 {color: maroon ;}

Whereas, colour names, or hexadecimal colour values or RGB or RGBA could also be used

Competency Level 10.6: Uses an authoring tool to create web pages

Learning Outcomes:

- Briefly explains web authoring tools
- Creates web pages using a web authoring tool

An authoring tool to create web pages

Any software or collection of software components that authors can be used to create or modify web content,to use by other people is called an authoring tool. Web authoring is the practice of creating web documents using modern web authoring software and tools. Web authoring software is a type of desktop publishing tool that allows users to navigate the tricky environment of HTML and web coding by offering a different kind of graphical user interface.

Examples of web authoring tools include:

- web page authoring tools (e.g., WYSIWYG HTML editors)
- software for directly editing source code or markup
- software for converting to web content technologies (e.g., "Save as HTML" features in office suites)
- integrated development environments (e.g., for web application development)
- software that generates web content on the basis of templates, scripts, command-line input or "wizard"-type processes
- software for generating/managing entire web sites (e.g., content management systems, courseware tools, content aggregators)
- email clients that send messages in web content technologies
- multimedia authoring tools
- debugging tools for web content
- software for creating mobile web applications
- scripting libraries
- web application frameworks, IDEs, and SDKs